# Fin500J Mathematical Foundations in Finance

# Topic 2: Matrix Calculus

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*Reference: Matrix Calculus, appendix from Introduction to Finite Element Methods book*

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# Outline

- The Derivatives of Vector Functions
- The Chain Rule for Vector Functions

# 1 The Derivatives of Vector Functions

Let $\mathbf{x}$ and $\mathbf{y}$ be vectors of orders $n$ and $m$ respectively:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix},$$

where each component $y_i$ may be a function of all the $x_j$, a fact represented by saying that $\mathbf{y}$ is a function of $\mathbf{x}$, or

$$\mathbf{y} = \mathbf{y}(\mathbf{x}).$$

If $n = 1$, $\mathbf{x}$ reduces to a scalar, which we call $x$. If $m = 1$, $\mathbf{y}$ reduces to a scalar, which we call $y$. Various applications are studied in the following subsections.

# 1.1 Derivative of Vector with Respect to Vector

The derivative of the vector $\mathbf{y}$ with respect to vector $\mathbf{x}$ is the $n \times m$ matrix

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \stackrel{\text{def}}{=} \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} & \cdots & \frac{\partial y_m}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_m}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_1}{\partial x_n} & \frac{\partial y_2}{\partial x_n} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

## 1.2 Derivative of a Scalar with Respect to Vector

If y is a scalar

$$\frac{\partial y}{\partial \mathbf{x}} \stackrel{\text{def}}{=} \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \vdots \\ \frac{\partial y}{\partial x_n} \end{bmatrix}.$$

It is also called the gradient of y with respect to a vector variable x, denoted by $\nabla y$ .

## 1.3 Derivative of Vector with Respect to Scalar

$$\frac{\partial \mathbf{y}}{\partial x} \stackrel{\text{def}}{=} \left[ \frac{\partial y_1}{\partial x} \quad \frac{\partial y_2}{\partial x} \quad \cdots \quad \frac{\partial y_m}{\partial x} \right]$$

# Example 1

Given $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

and

$$y_1 = x_1^2 - x_2$$
$$y_2 = x_3^2 + 3x_2$$

the partial derivative matrix $\partial \mathbf{y} / \partial \mathbf{x}$ is computed as follows:

$$\boxed{\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} \\ \frac{\partial y_1}{\partial x_3} & \frac{\partial y_2}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 & 0 \\ -1 & 3 \\ 0 & 2x_3 \end{bmatrix}}$$

# In Matlab

```
>> syms x1 x2 x3 real;
>> y1=x1^2-x2;
>> y2=x3^2+3*x2;
>> J = jacobian([y1;y2], [x1 x2 x3])
J =
[ 2*x1,   -1,    0]
[    0,    3, 2*x3]
```

**Note: Matlab defines the derivatives as the transposes of those given in this lecture.**

```
>> J'
ans =
[ 2*x1,    0]
[   -1,    3]
[    0, 2*x3]
```

# Some useful vector derivative formulas

$$\begin{bmatrix} C_{11} & C_{12} & \dots & C_{1n} \\ C_{21} & C_{22} & \dots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \dots & C_{nn} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^{n} x_t C_{1t} \\ \sum_{t=1}^{n} x_t C_{2t} \\ \vdots \\ \sum_{t=1}^{n} x_t C_{nt} \end{bmatrix}$$

$$\frac{\partial \; Cx}{\partial \mathbf{x}} = C^T$$

$$\frac{\partial \; x^T C}{\partial \mathbf{x}} = C$$

$$\frac{\partial \; x^T x}{\partial \mathbf{x}} = 2x$$

Homework

$$\frac{\partial \, \mathrm{Cx}}{\partial \mathrm{x}} = \begin{pmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{pmatrix} = C^T$$

# Important Property of Quadratic Form $\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x}$

$$\frac{\partial\ (\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x})}{\partial \mathbf{x}} = \left(\mathbf{C}+\mathbf{C}^{\mathrm{T}}\right)\mathbf{x}$$

Proof:

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^{n} x_t C_{1t} \\ \sum_{t=1}^{n} x_t C_{2t} \\ \vdots \\ \sum_{t=1}^{n} x_t C_{nt} \end{bmatrix}$$

$$\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = \sum_{i=1}^{n}\left[x_i \sum_{j=1}^{n}\left(x_j C_{ij}\right)\right]$$

$$\Rightarrow \frac{\partial(\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x})}{\partial x_k} = \frac{\partial\left\{\sum_{i=1}^{n}\left[x_i \sum_{j=1}^{n}\left(x_j C_{ij}\right)\right]\right\}}{\partial x_k} = \frac{\partial\left\{x_k \sum_{j=1}^{n}\left(x_j C_{kj}\right)\right\}}{\partial x_k} + \frac{\partial\left\{\sum_{i=1}^{n}\left[x_i x_k C_{ik}\right]\right\}}{\partial x_k}$$

$$= \sum_{j=1}^{n} x_j C_{kj} + \sum_{i=1}^{n} x_i C_{ik}$$

$$\Rightarrow \frac{\partial(\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x})}{\partial x} = \mathbf{C}\mathbf{x} + \mathbf{C}^{\mathrm{T}}\mathbf{x} = \left(\mathbf{C}+\mathbf{C}^{\mathrm{T}}\right)\mathbf{x}$$

If C is symmetric, $\dfrac{\partial\ (\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x})}{\partial \mathbf{x}} = 2C\,\mathbf{x}$

# 2 The Chain Rule for Vector Functions

Let

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix} \quad \text{and} \quad \mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{bmatrix}$$

where z is a function of y, which is in turn a function of x, we can write

$$\left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}}\right)^T = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \cdots & \frac{\partial z_1}{\partial x_n} \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \cdots & \frac{\partial z_2}{\partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial z_m}{\partial x_1} & \frac{\partial z_m}{\partial x_2} & \cdots & \frac{\partial z_m}{\partial x_n} \end{bmatrix}$$

Each entry of this matrix may be expanded as

$$\frac{\partial z_i}{\partial x_j} = \sum_{q=1}^{r} \frac{\partial z_i}{\partial y_q} \frac{\partial y_q}{\partial x_j} \qquad \begin{cases} i = 1, 2, \ldots, m \\ j = 1, 2, \ldots, n. \end{cases}$$

# The Chain Rule for Vector Functions (Cont.)

Then

$$\left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}}\right)^T = \begin{bmatrix} \sum \frac{\partial z_1}{y_q}\frac{\partial y_q}{\partial x_1} & \sum \frac{\partial z_1}{\partial y_q}\frac{\partial y_q}{\partial x_2} & \cdots & \sum \frac{\partial z_2}{\partial y_q}\frac{\partial y_q}{\partial x_n} \\ \sum \frac{\partial z_2}{\partial y_q}\frac{\partial y_q}{\partial x_1} & \sum \frac{\partial z_2}{\partial y_q}\frac{\partial y_q}{\partial x_2} & \cdots & \sum \frac{\partial z_2}{\partial y_q}\frac{\partial y_q}{\partial x_n} \\ \vdots & & & \\ \sum \frac{\partial z_m}{\partial y_q}\frac{\partial y_q}{\partial x_1} & \sum \frac{\partial z_m}{\partial y_q}\frac{\partial y_q}{\partial x_2} & \cdots & \sum \frac{\partial z_m}{\partial y_q}\frac{\partial y_q}{\partial x_n} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial z_1}{\partial y_1} & \frac{\partial z_1}{\partial y_2} & \cdots & \frac{\partial z_1}{\partial y_r} \\ \frac{\partial z_2}{\partial y_1} & \frac{\partial z_2}{\partial y_2} & \cdots & \frac{\partial z_2}{\partial y_r} \\ \vdots & & & \\ \frac{\partial z_m}{\partial y_1} & \frac{\partial z_m}{\partial y_2} & \cdots & \frac{\partial z_m}{\partial y_r} \end{bmatrix} \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_2}{\partial x_n} \\ \vdots & & & \\ \frac{\partial y_r}{\partial x_1} & \frac{\partial y_r}{\partial x_2} & \cdots & \frac{\partial y_r}{\partial x_n} \end{bmatrix}$$

$$= \left(\frac{\partial \mathbf{z}}{\partial \mathbf{y}}\right)^T \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\right)^T = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\frac{\partial \mathbf{z}}{\partial \mathbf{y}}\right)^T.$$

On transposing both sides, we finally obt $\frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{x}}\frac{\partial \mathbf{z}}{\partial \mathbf{y}},$

This is the chain rule for vectors (different from the conventional chain rule of calculus, the chain of matrices builds toward the left)

# Example 2

x, y are as in Example 1 and z is a function of y defined as

$$z = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{pmatrix}, \textit{and} \quad \begin{cases} z_1 = y_1^{\,2} - 2y_2 \\ z_2 = y_2^2 - y_1 \\ z_3 = y_1^2 + y_2^2 \\ z_4 = 2y_1 + y_2 \end{cases}, \text{ we have}$$

$$\frac{\partial z}{\partial y} = \begin{pmatrix} \frac{\partial z_1}{\partial y_1} & \frac{\partial z_2}{\partial y_1} & \frac{\partial z_3}{\partial y_1} & \frac{\partial z_4}{\partial y_1} \\ \frac{\partial z_1}{\partial y_2} & \frac{\partial z_2}{\partial y_2} & \frac{\partial z_3}{\partial y_2} & \frac{\partial z_4}{\partial y_2} \end{pmatrix} = \begin{pmatrix} 2y_1 & -1 & 2y_1 & 2 \\ -2 & 2y_2 & 2y_2 & 1 \end{pmatrix}.$$

Therefore,

$$\frac{\partial z}{\partial x} = \frac{\partial y}{\partial x}\frac{\partial z}{\partial y} = \begin{pmatrix} 2x_1 & 0 \\ -1 & 3 \\ 0 & 2x_3 \end{pmatrix} \begin{pmatrix} 2y_1 & -1 & 2y_1 & 2 \\ -2 & 2y_2 & 2y_2 & 1 \end{pmatrix} = \begin{pmatrix} 4x_1y_1 & -2x_1 & 4x_1y_1 & 4x_1 \\ -2y_1 - 6 & 1 + 6y_2 & -2y_2 + 6y_2 & 1 \\ -4x_3 & 4x_3y_2 & 4x_3y_2 & 2x_3 \end{pmatrix}$$

# In Matlab

```
>> z1=y1^2-2*y2;
>> z2=y2^2-y1;
>> z3=y1^2+y2^2;
>> z4=2*y1+y2;
>> Jzx=jacobian([z1; z2; z3; z4],[x1 x2 x3])
Jzx =
[       4*(x1^2-x2)*x1,        -2*x1^2+2*x2-6,                 -4*x3]
[               -2*x1,        6*x3^2+18*x2+1,      4*(x3^2+3*x2)*x3]
[       4*(x1^2-x2)*x1, -2*x1^2+20*x2+6*x3^2,      4*(x3^2+3*x2)*x3]
[                4*x1,                     1,                  2*x3]
>> Jzx'
ans =
[ 4*(x1^2-x2)*x1,    -2*x1,       4*(x1^2-x2)*x1,      4*x1]
[ -2*x1^2+2*x2-6,   6*x3^2+18*x2+1, -2*x1^2+20*x2+6*x3^2,  1]
[  -4*x3,           4*(x3^2+3*x2)*x3,  4*(x3^2+3*x2)*x3,  2*x3]
```